

Apurva Mandalika

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SUMMARY

Data Scientist with 5+ years of experience building machine learning, NLP, retrieval, AR and analytics solutions across customer behavior modeling, model monitoring, RAG systems, and AI-powered automation. Skilled in Python, SQL, PyTorch, Sentence Transformers, hybrid retrieval, LLM evaluation, and production-style AI applications, with recent hands-on work in agentic AI workflows, document Q&A, and business decision-support systems.

SKILLS

Python, SQL, pandas, NumPy, scikit-learn, XGBoost, PyTorch, forecasting, demand forecasting, time-series features, lag features, rolling-window features, predictive modeling, feature engineering, EDA, data cleaning, data mining, statistical analysis, A/B testing, model evaluation, model performance monitoring, customer analytics, KPI reporting, Tableau, Excel, Hive, Hadoop, FastAPI, Flask, Streamlit, Docker, AWS, GCP, Git, GitHub, Gemini API, Claude API, RAG, prompt engineering, LLM routing, semantic search, hybrid retrieval, vector search, reranking, conversational document Q&A, AI copilots.

EXPERIENCE

Senior Consultant – Data Scientist, Heartland Community Network

Sep 2025 – Aug 2026

- Built and deployed a Claude API-powered document Q&A assistant on Streamlit Cloud, enabling grounded summaries and plain-English answers from uploaded business documents, and achieved approximately 91% accuracy across 20+ evaluation pairs.
- Implemented a customer segmentation model for a retail client using K-Means clustering, identifying 4 distinct buyer personas and increasing targeted campaign ROI by 22%.

Senior Data Scientist, [24]7.ai

Jun 2022 – Jul 2023

- Developed AR-based customer-support applications across customer-facing iOS, agent-facing iOS, and web interfaces, integrating real-time camera and video data into guided and live-agent support workflows.
- Built and evaluated computer-vision and AR interaction prototypes using Swift and ARKit, testing visual overlays across different devices, environments, camera orientations, and user scenarios.
- Collaborated with platform, product, and QA teams to transition AR video-support prototypes to the company's GCP-based production platform.
- Led 15+ proof-of-concept implementations for video-based AR customer-support capability that improved resolution rates by approximately 50% during internal testing and resulted in a published patent PCT/IB2023/050635.

Data Scientist, [24]7.ai

Jul 2020 – May 2022

- Introduced and implemented AR solutions using ARKit and computer vision to develop video-based customer support.
- Performed customer analytics reporting, generating ad hoc reports in Excel for stakeholders, delivering 10+ reports per month.
- Built and maintained a model-performance dashboard using Hive, SQL, Excel, and Python to monitor KPIs, performance trends, and business impact across 23 client predictive-targeting models, increasing actionable insights by 40%.
- Created comprehensive model documentation in JIRA and Confluence, improving transparency and accelerating team onboarding.

Analytics Consultant, [24]7.ai

Jan 2019 – Jun 2020

- Created Time on Page (TOP) and Page-Level Propensity to Purchase after Chat (P2PC) models, increasing chat propensity by 8%, RPC by 6%, and conversion rates by 12%.
- Deployed scalable predictive targeting models for 4 clients using Hadoop and FlashML for customer behavior modeling.
- Conducted detailed EDA, preprocessing, anomaly filtering, and A/B testing, improving model performance by nearly 5%.

PROJECTS

Walmart Demand Forecasting Copilot | Tools: Python, Forecasting, LLM Routing, FastAPI, XGBoost, Streamlit.

- Developed an AI-powered retail analytics assistant using Gemini-based routing to invoke XGBoost forecasting and KPI tools, with lag/rolling features served through FastAPI and Streamlit for interactive demand planning.

Multimodal Classification Model | Tools: Python, PyTorch, CNNs, Neural Networks

- Developed a multimodal fusion model combining CNN-based image features with neural-network-based audio features for multimodal MNIST classification, achieving 98.92% validation accuracy.

RetailIQ AI: Retail Knowledge Assistant | Tools: Python, RAG, FAISS, BM25, cross-encoder, Gemini API, FastAPI.

- Built a production-style retail RAG assistant using FAISS, BM25, and cross-encoder reranking for grounded answers, achieving 100% source retrieval and ~89% answer coverage across 15+ QA pairs.

Agentic RAG System with MCP Tool Orchestration | Tools: Python, Anthropic MCP, Claude API, FAISS, BM25, FastAPI.

- Built an agentic RAG application where a Claude-powered agent routes user queries across hybrid retrieval, web search, and structured data lookup tools using Anthropic MCP, FAISS, BM25, cross-encoder reranking, and FastAPI.
- Evaluated the system across a 25-document corpus and 15 test queries, achieving 93%+ retrieval coverage with multi-source reasoning and source attribution in every response.

EDUCATION

Texas A&M University, Master's in Computer Science (CGPA: 3.9/4)

Aug 2023 – May 2025

Amrita Vishwa Vidyapeetham, BTech in Computer Science and Engineering (CGPA: 9.35/10, Dept. 8th rank)

Aug 2015 – May 2019